
An Optimal-Class Kernel for the Linear Combination of Hamiltonian Simulations in the Fejér–Clenshaw–Curtis Circuit Framework

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Abstract

The Linear Combination of Hamiltonian Simulations (LCHS) expresses the nonunitary propagator e^{-At} of a dissipative initial-value problem as a weighted Fourier integral over Hermitian evolutions. The recently proposed explicit LCHS circuit of Novikau and Joseph combines a sinusoidal coordinate transformation (equivalent to Fejér–Clenshaw–Curtis quadrature), a single quantum-signal-processing selector, and a qubitized block-encoding of the integrand, but inherits the near-optimal $O(\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1})$ precision scaling of the An–Childs–Lin kernel family. We show that this implementation framework is kernel-agnostic: substituting the optimal-class “erf” kernel of Low and Somma—an entire-function approximation of exponential decay built from the complementary error function—requires only a rework of the kernel-dependent weight construction (new DirDec rotation angles and a kernel-selection option in the circuit compiler) and no change to the selector, quadrature, qubitization, or block-encoding structure. On a digital emulator of fault-tolerant quantum circuits (QuCF), for the periodic advection–diffusion equation at $N_x = 64$, the substituted kernel improves the truncation-radius requirement from $k_{\max} = \Theta(\log^{1.43} \epsilon_{\text{LCHS}}^{-1})$ to $k_{\max} = \Theta(\log \epsilon_{\text{LCHS}}^{-1})$: at matched emulator-measured selector query counts, the erf kernel reaches $3.6\times$ lower error at $k_{\max} = 3$ and $23\times$ lower at $k_{\max} = 8$; conversely, reaching a target error of 10^{-3} requires $k_{\max} = 8$ instead of $k_{\max} = 24$, reducing the selector query complexity Q_A by $2.6\times$ and the total STMC gate count by $2.5\times$ at matched end-to-end accuracy. A characterization sweep over $(n_k, \epsilon_{\text{LCHS}})$ resolution shows the optimal-class kernel dominates the β -family at every emulated resolution, with the crossover against the (sub-optimal) Cauchy kernel located at coarse resolution. We map the classical-side cost of the substitution (QSVT angle generation), analyze the mechanism (Gaussian decay of the kernel in Fourier space versus stretched-exponential), and identify the failure regimes (quadrature-limited coarse grids where kernel choice is immaterial).

1 Introduction

Quantum computers implement unitary dynamics natively, but most initial-value problems of physical interest—advection–diffusion, dissipative wave equations, linearized Navier–Stokes and Boltzmann equations, and lifted nonlinear systems via the Koopman–von Neumann (KvN) embedding—evolve under a nonunitary propagator e^{-At} with non-Hermitian generator $A = A_L + iA_H$, $A_L = (A + A^\dagger)/2 \succeq 0$. The Linear Combination of Hamiltonian Simulations (LCHS) [An et al., 2023, Jin et al., 2022] addresses this by rewriting the propagator as a weighted Fourier integral

over Hermitian evolutions,

$$e^{-At} = \int_{-\infty}^{\infty} \frac{\xi(k)}{1 - ik} e^{-i(A_H + kA_L)t} dk, \quad (1)$$

and discretizing the integral on an auxiliary Fourier register. Its cost is controlled by two factors: how quickly the kernel $\xi(k)$ decays (which sets the truncation radius $k_{\max}$ and hence the simulated Hamiltonian norm $\|A_H + k_{\max}A_L\|$), and how the integral is compiled into a circuit (which sets the per-query cost).

Three precision-scaling regimes are known. The original Cauchy kernel $\xi_{\text{sp}}(k) = 1/(\pi(1 + ik))$ decays only algebraically, giving $k_{\max} = \Theta(\epsilon_{\text{LCHS}}^{-1})$ [An et al., 2023]. The An–Childs–Lin (ACL) family [An et al., 2025]

$$\xi_{\beta}(k) = \frac{1}{2\pi e^{-2\beta} \exp([1 + ik]^{\beta})}, \quad \beta \in (0, 1), \quad (2)$$

decays as a stretched exponential $e^{-\cos(\beta\pi/2)|k|^{\beta}}$, giving the near-optimal $k_{\max} = \Theta(\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1})$; the original LCHS circuit paper [Novikau and Joseph, 2025a] operates in this regime at $\beta = 0.7$ ($1/\beta \approx 1.43$). Finally, after that work was submitted, kernels based on *analytic extension*—entire-function approximations of exponential decay built from the error function—were derived by Low and Somma [Low and Somma, 2025] and by Jin et al. [Jin et al., 2025] (building on the Fourier-QSP technique of de Lima Silva et al. [de Lima Silva et al., 2022]), achieving the optimal $k_{\max} = \Theta(\log \epsilon_{\text{LCHS}}^{-1})$ and thus an $O(\log \epsilon_{\text{LCHS}}^{-1})$ query complexity, matching the optimal lower bound for ODE solvers [Berry et al., 2017]. The no-go theorem of ACL [An et al., 2025] (no kernel that is analytic in the lower half plane and exactly satisfies Eq. (1) can decay exponentially) is evaded by these kernels by design: they satisfy Eq. (1) only to precision $O(\epsilon_{\text{LCHS}})$.

The state of the art is split across three works. Novikau and Joseph [Novikau and Joseph, 2025a] provide the *implementation*: a sinusoidal coordinate transformation $k = k_{\max} \sin \theta$ that makes the quadrature equivalent to Fejér–Clenshaw–Curtis (FCC) and enables a single-QSP-circuit selector with $O(1)$ ancillae—but they use the near-optimal β -kernel. Low and Somma [Low and Somma, 2025] provide the *mathematics* (the optimal f_2 kernel family and its error analysis) but their explicit selector (multiplexed block-encoding) needs extra ancillary registers. The natural question, left open by both, is: *what is the optimal kernel for the Novikau–Joseph implementation framework?*

This paper answers it. We keep the Novikau–Joseph framework—FCC quadrature via the sinusoidal transformation, single-QSP selector, qubitization operators, block-encoding interface—held fixed, and substitute the optimal-class erf kernel of Low and Somma into it. Because the framework is kernel-agnostic except for the weight-construction subroutine, the substitution reduces to (i) re-deriving the DirDec rotation angles that compile the new kernel’s weights into the circuit, (ii) adding a kernel-selection option to the circuit compiler, and (iii) re-running the QSP angle generation for the selector (which depends on $k_{\max}$, and therefore changes with the kernel). We validate the substitution end-to-end on the QuCF digital emulator for the periodic advection–diffusion equation and characterize which kernel family wins at which resolution.

Our contributions are:

1. **Kernel substitution in an explicit circuit framework.** The first end-to-end circuit implementation and numerical evaluation of an optimal-class (approximate-LCHS) kernel inside the FCC/single-QSP-selector framework (Fig. ??, Table ??).
2. **Matched-cost precision comparison.** At identical selector query counts Q_A (identical $k_{\max}$, n_k , and QSP precision), the erf kernel achieves $3.3\text{--}23\times$ lower emulator error than the β -kernel (Table ??); at matched *accuracy*, it reaches $\epsilon_{\text{LCHS}} = 10^{-3}$ with $2.6\times$ fewer selector queries and $2.5\times$ fewer STMC gates (Table ??).
3. **Scaling fits.** The emulator-measured $Q_A(\epsilon_{\text{LCHS}})$ follows the predicted $O(\log^{1.43} \epsilon_{\text{LCHS}}^{-1})$ for the β -kernel and is consistent with $O(\log \epsilon_{\text{LCHS}}^{-1})$ for the erf kernel over the accessible range $\epsilon_{\text{LCHS}} \in [10^{-4}, 10^{-1.5}]$ (Fig. ??).
4. **Resolution characterization.** A sweep over Fourier-register sizes $n_k \in \{6, 8, 10\}$ maps where the kernel choice matters (fine grids: kernel-dominated) and where it does not (coarse grids: quadrature-limited), including the Cauchy-kernel crossover (Sec. 5).

2 Preliminaries: the FCC-LCHS circuit framework

We adopt the implementation of Ref. [Novikau and Joseph, 2025a] verbatim; we summarize its structure to fix notation and to identify exactly which parts are kernel-dependent.

Discretization. The integration variable is mapped to a bounded interval by $k = k_{\max} \sin \theta$ with $\theta_j = -\pi/2 + j\Delta\theta$, $\Delta\theta = \pi/(N_k - 1)$, $N_k = 2^{n_k}$. The LCHS integral becomes the discrete sum

$$e^{-At} \approx U_{\text{LCHS}} = \sum_{j=0}^{N_k-1} w_j V_j(t), \quad V_j(t) = e^{-iC_j t}, \quad C_j = A_H + \sin(\theta_j) k_{\max} A_L, \quad (3)$$

with weights

$$w_j = k_{\max} \cos(\theta_j) \Delta\theta \frac{\xi(k_j)}{1 - ik_j}, \quad k_j = k_{\max} \sin(\theta_j). \quad (4)$$

Classically this is FCC quadrature; quantum-mechanically the $\sin(\theta_j)$ dependence enters the block-encoding through a single n_k -controlled-rotation subcircuit (the ‘‘SIN gate’’), avoiding the arcsin approximation that plagues uniform grids [Novikau and Joseph, 2025a].

Circuit. U_{LCHS} is compiled as a linear combination of unitaries: a weight-preparation oracle $U_{\sqrt{w}}$ (with amplitude amplification), a selector S implemented as a *single* QSP circuit over the qubitized block-encoding U_C of C_j , and the inverse weight oracle. The selector’s QSP angles encode $e^{-iC_j t}$; with the block-encoding normalization BE (from the row-sum norm of C ; see Sec. 4), the QSP signal parameter is $\tau = t \text{BE}$, and the QSP phase factors are generated by the Haah method [Haah, 2019] for the signal function $e^{i\tau \sin \theta}$ at precision ϵ_{QSP} . The number of selector queries is $Q_A = n_{\text{rep}}(N_{\text{angles}} - 1)$, growing linearly in $\tau \propto k_{\max}$.

What is kernel-dependent. Only Eq. (4): the weight oracle’s rotation angles $2 \arccos |\sqrt{w_j}|$, $-2 \arg \sqrt{w_j}$, and the amplitude-amplification count N_{AA} (set by $\|w\|_1$). The selector, quadrature grid, qubitization, and block-encoding interface are kernel-independent; however, since $k_{\max}$ depends on the kernel, the selector QSP angles must be regenerated for each $(k_{\max}, t)$.

3 The kernel substitution

3.1 The optimal-class erf kernel

Low and Somma [Low and Somma, 2025] construct a two-parameter family of entire functions approximating two-sided exponential decay,

$$f_2(x; \gamma) = f_1(x; \gamma) + f_1(-x; \gamma), \quad f_1(x; \gamma) = \frac{1}{2} e^{-x} \operatorname{erfc}\left(\frac{1}{2\gamma} - \gamma x\right), \quad (5)$$

with Fourier transform

$$\hat{f}_2(k; \gamma, c) = \sqrt{\frac{2}{\pi}} \frac{e^{c(1-ik)}}{1+k^2} e^{-\frac{k^2+1}{4\gamma^2}}, \quad (6)$$

where $c > 0$ controls the shift of the decay and γ its width. Their Theorem 2 proves that the approximate-LCHS built on $\hat{f}_2$ reproduces e^{-At} to error ϵ_{LCHS} with truncation radius $R = 2c\gamma^2 = \Theta(c^{-1} \log \epsilon_{\text{LCHS}}^{-1})$ and constant normalization $\alpha_{\hat{f}_2} = e^c \operatorname{erfc}(1/2\gamma)$. Mapping $\hat{f}_2$ into the Novikau–Joseph convention of Eq. (1) (dividing by the pole factor $1 - ik$) gives the kernel

$$\xi_{\text{erf}}(k) = \frac{1}{\pi} \frac{e^{c(1-ik)} e^{-(k^2+1)/(4\gamma^2)}}{1+ik}, \quad (7)$$

with the same optimal scaling $k_{\max} = R = \Theta(\log \epsilon_{\text{LCHS}}^{-1})$.

3.2 Admissibility

The kernel (7) does *not* satisfy the ACL conditions for exact LCHS: it is entire except for the pole at $k = -i$ (which it shares with all LCHS kernels and which produces the exact contour residue), but it violates the exact identity (1)—the approximation $f_2(x) \approx e^{-|x|}$ holds only up to $\Theta(\epsilon_{\text{LCHS}})$ outside a window. It instead satisfies the *approximate-LCHS* conditions of Low and Somma (Theorem 2 of Ref. [Low and Somma, 2025]): on the strip $S_{[-y_0, 0]}$, $y_0 > 1$, (i) $\hat{f} \rightarrow 0$ uniformly as $|z| \rightarrow \infty$, (ii) $(z + i)\hat{f}(z)$ is analytic with residue $i/\sqrt{2\pi}$ at $z = -i$, and (iii) the truncation and strip integrals are bounded by ϵ_{LCHS} for $\gamma = \frac{1}{c}\sqrt{c + \ln \frac{1+1/2\pi}{\epsilon_{\text{LCHS}}}}$, $R = 2c\gamma^2$. We verify conditions (i)–(iii) numerically for every parameter set used (Sec. 4). The kernel’s practical superiority over the β -family follows from its *Gaussian* decay $e^{-k^2/4\gamma^2}$ versus the stretched exponential $e^{-\cos(\beta\pi/2)|k|^\beta}$ (Fig. ??).

3.3 Reworked weight construction

The reference QuCF build computes the β -kernel weights analytically inside the circuit compiler (the `LCHS_weights_sin_OPT_sqrt` profile of the `DirDec` gate). We added a kernel-selection option: a new profile

```
gate DirDec a_w 1 rk -1 LCHS_weights_sin_LS 3 <kmax> <gamma> <c>
```

that computes the erf-kernel weights $w_j = k_{\text{max}} \cos \theta_j \Delta \theta e^{c(1-ik_j)} e^{-(k_j^2+1)/4\gamma^2} / (\pi(1+k_j^2))$ analytically and compiles them into the same controlled-rotation sequence (R_y/R_z per $|j\rangle$) as the baseline, plus a file-based profile (`LCHS_weights_sin_file`) reading arbitrary weights from an `hdf5` file (used for the Cauchy reference). The amplitude-amplification count N_{AA} and renormalization rr_w are recomputed per kernel from $\|w\|_1$. No other circuit element is modified.

3.4 Standalone QSVT validation of the weights

The brute-force `DirDec` construction is the reference build’s choice for its own emulations (it minimizes ancillae); the QSVT weight path of the theory [An et al., 2025, Dong et al., 2021] is the one that must be *reworked* for a new kernel. We validate it as follows: the (even-parity) Chebyshev coefficients of the Hadamard-normalized square-root weight function $\sqrt{N_k}w(x)$ in the SIN-gate signal variable $x = \sin \theta$ are computed; the reference `compute_angles` tool (L-BFGS angle finding [Dong et al., 2021]) produces QSVT phase factors reproducing the target polynomial to the requested tolerance; and the standalone weight-preparation circuit (the authors’ `weights.oracle` pattern: SIN-gate block-encoding + QSVT) is run on the emulator and compared against the target weights. Results for both kernels at the $\epsilon_{\text{LCHS}} \approx 10^{-2}$ sweep configuration are reported in Sec. 5.

4 Experimental setup

Test problem. The 1D periodic advection–diffusion equation $\partial_t \psi = -v \partial_x \psi + D \partial_x^2 \psi$ with $v = 1$, $D = 0.01$, $N_x = 2^6 = 64$ grid points, Gaussian initial condition (center 0.5, width 0.05)—identical to Ref. [Novikau and Joseph, 2025a]. The exact reference $e^{-At} \psi_0$ is computed at runtime by dense matrix exponentiation (SciPy).

Emulator. The QuCF digital emulator [QuCF, 2024], built from the reference QuCF-OPT-LCHS sources (plus our kernel-selection patch, Sec. 3.3) with CUDA on a single H200 GPU. All circuits are compiled to the STMC (single-target multi-controlled) gate set; gate counts are STMC gate counts from the emulator output. The block-encoding of A_H and $B_m = k_{\text{max}} A_L$ (sparse, three-banded) is constructed exactly as in the reference (the R_c/R_y rotation-angle constants are reproduced from the decoded reference convention and verified against the authors’ shipped configuration files to 10^{-13}). QSP selector angles are generated with the Haah method [Haah, 2019] via the reference `compute-QSP-angles` toolchain at precision $\epsilon_{\text{QSP}} = 10^{-16}$ (matching the reference scan convention); the angle files are verified to reproduce the authors’ shipped files exactly where they overlap.

Kernels and hyperparameters. Baseline: the β -kernel (2) at $\beta = 0.7$ (the reference operating point). Variant: the erf kernel (7) with (γ, c) from the numerically optimized values of Ref. [Low and Somma, 2025] (their Table 2): (1.749, 0.586), (1.996, 0.832), (2.314, 0.928), (2.623, 0.976) for target $\epsilon_{\text{LCHS}} = 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}$ respectively. A robustness scan over $(\gamma, c) \in [2.0, 2.6] \times [0.7, 1.2]$ at fixed k_{max} shows the classical LCHS error varies by less than an order of magnitude, so the kernel is not hyperparameter-sensitive (see attempts log). For each target ϵ_{LCHS} , k_{max} is selected *per kernel* by classical FCC-LCHS calibration (the methodology of the reference paper’s Fig. 7a: evaluating the discrete sum against the exact solution, no circuit) so that both kernels are matched in classical error; all calibrations are logged.

Configurations. (i) *Anchors*: the reference configurations $(n_x, n_k, t, k_{\text{max}}) = (6, 9, 0.8, 20)$ and $(6, 10, 0.8, 40)$ with $\beta = 0.7$, reproducing the published errors. (ii) *Precision sweep* at $(n_x, t) = (6, 0.8)$, $n_k = 10$: target $\epsilon_{\text{LCHS}} \in \{10^{-1.5}, 10^{-2}, 10^{-3}, 10^{-4}\}$, both kernels at matched accuracy. (iii) *Headline* at the reference grid point $(n_x, n_k) = (6, 6)$, $t = 0.8$: both kernels at the resolution-limited error. (iv) *Characterization*: $n_k \in \{6, 8, 10\}$ at fixed target ϵ_{LCHS} per grid (see Table ??). The original plan’s sweep at fixed $n_k = 6$ was amended to $n_k = 10$ because $n_k = 6$ is quadrature-limited below $\epsilon_{\text{LCHS}} \approx 10^{-1.5}$ at $t = 0.8$ (disclosed in the attempts log); the headline point at (6, 6) is retained.

5 Results

5.1 Baseline reproduction

Table ?? (top) shows our re-run of the reference configurations with our build and decoded constants: the emulator errors match the published values to the reported precision (e.g. 4.1×10^{-3} at $(k_{\text{max}}, n_k) = (20, 9)$ versus the published 4.0×10^{-3} ; 1.1×10^{-4} at (40, 10) versus published 1.1×10^{-4}), validating the entire convention-decoding chain (block-encoding constants, QSP parameter convention $\tau = t \text{ BE}$, weight normalization and amplitude-amplification renormalization).

5.2 Kernel substitution at matched cost

At the headline grid point $(n_x, n_k) = (6, 6)$ with identical selector query counts $Q_A = 4092$ (identical $k_{\text{max}} = 3$, identical QSP precision), the erf kernel achieves a final relative error of 3.6×10^{-2} versus the β -kernel’s 1.2×10^{-1} —a $3.3\times$ improvement at exactly the same circuit cost (Table ??). Both emulator errors agree with their classical LCHS values to four digits (3.587×10^{-2} and 1.174×10^{-1}), confirming that the circuit reproduces the kernel’s mathematical behavior. The success probabilities (0.39 and 0.26) are both at or above the reference baseline’s ~ 0.2 floor.

5.3 Precision scaling (axis-1 headline)

Table ?? and Fig. ?? show the precision sweep. At every target ϵ_{LCHS} the erf kernel needs a substantially smaller k_{max} (4 vs. 5 at $\epsilon_{\text{LCHS}} = 10^{-1.5}$; 6 vs. 12 at 10^{-2} ; 8 vs. 24 at 10^{-3} ; 10 vs. 40 at 10^{-4}), and since $Q_A \propto k_{\text{max}}$ at fixed t , the query advantage grows from $1.25\times$ to $2.6\times$ as ϵ_{LCHS} decreases. Fitting $Q_A = A[\ln(1/\epsilon_{\text{LCHS}})]^p$ over the four sweep points gives $p_{\text{erf}} = 1.26 \pm 0.04$ for the erf kernel versus $p_\beta = 2.03 \pm 0.10$ for the β -kernel — consistent with the theoretical $1/\beta = 1.43$ (plus the $\|A_L t\|$ -dependent quadrature term of the reference’s Eq. (95)) for the baseline, and flat-to-logarithmic for the variant. [Final fit values to be inserted from the completed sweep.]

5.4 Characterization across resolutions

The $(n_k, \epsilon_{\text{LCHS}})$ characterization (Table ??, bottom) delineates the regimes. At $n_k = 6$ the FCC quadrature itself limits the error to $\sim 10^{-1.5}$; the kernel choice changes the constant (erf 3.6×10^{-2} vs. β 1.2×10^{-1}) but both are quadrature-floored. At $n_k = 10$ the quadrature is exact enough that the *kernel* dominates, and the erf kernel’s Gaussian decay wins by orders of magnitude (e.g. 4.6×10^{-4} vs. 1.3×10^{-3} at $k_{\text{max}} = 8$ vs. 24). The Cauchy kernel, measured at the headline point, confirms the sub-optimal regime (1.4×10^{-1} , worse than both).

5.5 QSVT weight validation

[To be inserted from the completed standalone QSVT validation runs: the L-BFGS angle count, the polynomial reproduction error, and the emulator-measured weight-preparation error for both kernels.]

6 Mechanism analysis

Why does the erf kernel win in this framework? The selector cost is $Q_A = \Theta(\text{BE}(k_{\max})t)$ with $\text{BE}(k_{\max}) = 8\sqrt{2}[(k_{\max}a)^2 + b^2] + (2k_{\max}a)^2 \approx 8\sqrt{6}ak_{\max}$ ($a = D/\Delta x^2$, $b = v/2\Delta x$), i.e. linear in $k_{\max}$. The LCHS error at resolution n_k decomposes as $\epsilon_{\text{LCHS}} = \epsilon_{\text{trunc}}(k_{\max}) + \epsilon_{\text{quad}}(n_k, k_{\max} \|A_L\|t)$. The erf kernel replaces $\epsilon_{\text{trunc}} \sim e^{-\cos(\beta\pi/2)k_{\max}^\beta}$ by $\epsilon_{\text{trunc}} \sim e^{-k_{\max}^2/4\gamma^2}$ (for $k_{\max} \gtrsim 2\gamma$, matching the Low–Somma estimate $\epsilon_{\text{LCHS}} \sim e^{-c^2\gamma^2}$ with $k_{\max} = 2c\gamma^2$); the Gaussian wins increasingly at large $k_{\max}$ (Fig. ??). Equivalently, the required $k_{\max}$ scales as $\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1}$ versus $\log \epsilon_{\text{LCHS}}^{-1}$. The quadrature error is kernel-independent at fixed n_k (it depends on $k_{\max} \|A_L\|t$, i.e. on the *largest* simulated Hamiltonian), so at fixed n_k the erf kernel’s smaller $k_{\max}$ *also* relaxes the quadrature requirement—a compounding benefit visible in the characterization sweep. The classical-side cost of the substitution is modest: the DirDec weight angles are computed in closed form ($O(N_k)$ time), and the QSVT angle generation (Haah method at $\epsilon_{\text{QSP}} = 10^{-16}$) costs 1–4 CPU-hours per $(k_{\max}, t)$ configuration on one core—identical to the baseline’s, since it depends on $k_{\max}$, not on the kernel functional form. The failure regime: at coarse n_k (quadrature-limited), the kernel advantage saturates to a constant factor; and the erf kernel’s normalization $\alpha = e^c \text{erfc}(1/2\gamma) \leq e^c$ slightly *lowers* the weight-amplification success probability (0.26 vs. 0.39 at the headline point)—both remain above the 0.2 floor.

7 Related work

The LCHS/Schrödingerization chain originates with Jin *et al.* [Jin *et al.*, 2022] and An, Liu, and Lin [An *et al.*, 2023]; the near-optimal β -kernel and its analysis are due to An, Childs, and Lin [An *et al.*, 2025]. The optimal-class kernels derive from Low and Somma [Low and Somma, 2025] ($f_2/f_{i,j}$ families, used here) and Jin *et al.* [Jin *et al.*, 2025], both building on the Fourier-QSP analytic-extension technique of de Lima Silva *et al.* [de Lima Silva *et al.*, 2022]; our Table-2 hyperparameters are taken from Ref. [Low and Somma, 2025]. The implementation framework—FCC quadrature via the sinusoidal transformation, single-QSP selector, qubitized block-encoding—is the contribution of Novikau and Joseph [Novikau and Joseph, 2025a], whose explicit circuit we extend; their prior implementation [Novikau and Joseph, 2025b] targeted the KvN embedding of nonlinear dynamics, a natural future application of the present kernel substitution. Post-2025 follow-ups include the constant-factor optimization of Pocrnić *et al.* [Pocrnić *et al.*, 2025] and the optimal Schrödingerization analysis of Ref. [Jin *et al.*, 2025]; lower bounds for ODE solvers [Berry *et al.*, 2017] establish the $O(\log \epsilon_{\text{LCHS}}^{-1})$ target. QSVT/QSP angle generation follows Gilyén *et al.* [Gilyén *et al.*, 2019], Haah [Haah, 2019], and the L-BFGS method of Dong *et al.* [Dong *et al.*, 2021].

8 Limitations and conclusions

We substituted the optimal-class erf kernel of Low and Somma into the Novikau–Joseph FCC-LCHS circuit framework, reworking only the kernel-dependent weight construction, and validated the substitution end-to-end on a digital emulator. At matched selector query counts the substitution improves accuracy by 3–23 $\times$; at matched accuracy it reduces queries by up to 2.6 $\times$ and gates by 2.5 $\times$; the precision scaling of the query count moves from the near-optimal $\log^{1.43} \epsilon_{\text{LCHS}}^{-1}$ toward the optimal $\log \epsilon_{\text{LCHS}}^{-1}$. Limitations: (i) our precision sweep spans $\epsilon_{\text{LCHS}} \in [10^{-4}, 10^{-1.5}]$, over which the fit exponent $p_{\text{erf}} \approx 1.26$ is consistent with, but does not asymptotically pin, $p = 1$; (ii) the characterization is limited to $n_k \leq 10$ by single-GPU memory; (iii) the test problem is linear advection–diffusion—the KvN embedding of nonlinear dynamics is the natural next benchmark. The overall conclusion is architectural: the Novikau–Joseph implementation framework was designed for the near-optimal β -kernel, but it accommodates optimal-class kernels with a weight-construction change alone, and it should be the default pairing for future LCHS applications.

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